

Hamed Daoud

Alexandria, Egypt

itshamedahmed@gmail.com
github.com/HamedDaoud

Phone: +201553455984
linkedin.com/in/hameddaoud

EDUCATION

- **Egypt-Japan University of Science and Technology (EJUST)** **Alexandria, Egypt**
B.Sc. in Computer Science & Information Technology — AI & Data Science Major
CGPA: 3.87 / 4.00
2022 – Fall 2026 (Expected)
- **University of Tsukuba** **Tsukuba, Japan**
School of Informatics — EJEP Short-Term Exchange Program
Sept 2025 – Feb 2026

EXPERIENCE

- **University of Tsukuba — Research Lab** **Tsukuba, Japan**
Research Intern, Deep Learning for Biological Signal Processing
Sept 2025 – Feb 2026
 - Implemented DDPM, DDIM, and LDM diffusion variants in PyTorch for EEG reconstruction on the anphy-sleep dataset (29 subjects, 83-channel HD-EEG).
 - Compared diffusion parameterizations (data-, noise-, and velocity-prediction); selected velocity-prediction as the optimal training objective.
 - Built EEG preprocessing pipelines covering signal segmentation and data augmentation to expand the training distribution.
 - Proposed signal-space diffusion over latent-space alternatives; adopted as my architecture after lab approval, tailored to EEG noise distributions.

PROJECTS

- **3D-Agent — Graduation Project** github.com/hecrw/3D-Agent
 - Collaborated with a team of 4 on a multi-model 3D generation pipeline (TRELLIS.2, PartCrafter, Hunyuan3D via Google AI Studio) backed by FastAPI + Modal inference; environment managed with uv.
 - Designed the Django web interface and integrated the agent runtime with a 12-tool layer — the project's primary user-facing surface for text-to-3D generation and editing.
 - Co-developed an image-restyling preprocessing stage that conforms generated 2D inputs to an Objaverse-style distribution, improving 3D reconstruction quality.
- **FraudShieldML — Credit Card Fraud Detection** github.com/HamedDaoud/FraudShieldML · live demo
 - Engineered an end-to-end fraud detection system on a severely imbalanced dataset (0.17% positives), reaching **0.93 precision, 0.82 recall, and 0.86 PR AUC**.
 - Built a feature pipeline with log-transformed amounts, cyclic temporal encoding, standard scaling, and SMOTE resampling to address class imbalance.
 - Threshold-tuned a Random Forest classifier; served real-time inference through a FastAPI service and a Gradio UI, containerized with Docker Compose and deployed on Hugging Face Spaces.
- **Falcon-RW 1B — 4-bit Quantized Text Generation** github.com/HamedDaoud/quantized-falcon-rw-1b · live demo
 - Quantized Falcon-RW 1B to 4-bit NormalFloat (NF4) with double quantization via **bitsandbytes**, reducing memory footprint by **approximately 75%** versus float32.
 - Configured **bf16** compute dtype with a CUDA → MPS → CPU device-fallback chain and cached loading, so the same model loader powers both the Gradio UI and the FastAPI service.
 - Containerized the application with Docker and deployed to Hugging Face Spaces; backed the Gradio interface with a FastAPI **/generate** endpoint for programmatic use.

TECHNICAL SKILLS

- **Programming:** Python, C++, JavaScript, SQL
- **ML & Deep Learning:** PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers, NumPy, Pandas, Matplotlib, XGBoost
- **AI Methods:** LLMs, NLP, Diffusion Models, Model Quantization, Computer Vision
- **MLOps & Backend:** Docker, Docker Compose, FastAPI, Modal, Hugging Face Spaces, Django, Gradio, MLflow, Optuna, Google AI Studio, Git, Linux, uv
- **Cloud:** AWS (S3, EC2, Lambda, IAM, CloudWatch, RDS)
- **Languages:** English (Native), Arabic (Native)